

COMP3026E Distributed Systems

15. Preview the Exam

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Question Set 1: True or False (2pt × 10)



1. In a consensus protocol like Paxos or Raft, a proposed value is considered “chosen” as soon as it has been accepted by a single acceptor or replica node.
2. A consistency model A is considered strictly stronger than a consistency model B if model A allows a strict subset of the behaviors that model B allows.



Question Set 1: True or False (2pt × 10)



1. In a consensus protocol like Paxos or Raft, a proposed value is considered “chosen” as soon as it has been accepted by a single acceptor or replica node.

False. In Paxos and Raft, a value is only considered “chosen” once it has been accepted by a simple majority of nodes.

2. A consistency model A is considered strictly stronger than a consistency model B if model A allows a strict subset of the behaviors that model B allows.



Question Set 1: True or False (2pt × 10)



1. In a consensus protocol like Paxos or Raft, a proposed value is considered “chosen” as soon as it has been accepted by a single acceptor or replica node.

False. In Paxos and Raft, a value is only considered “chosen” once it has been accepted by a simple majority of nodes.

2. A consistency model A is considered strictly stronger than a consistency model B if model A allows a strict subset of the behaviors that model B allows.

True. A consistency model is strictly stronger if its set of allowed behaviors is a strict subset of the weaker model’s behaviors.



Question Set 2: Multiple Choice Selection (3pt × 10)



1. In Cristian's Algorithm, a client sends a request at local time $T1 = 100$. The server receives the request at $T2 = 105$ and sends a response at $T3 = 110$. The client receives the response at $T4 = 125$. What is the round-trip time (δ) the client uses to estimate the clock offset?
- A. 15ms B. 20ms C. 25ms D. 30ms



Question Set 2: Multiple Choice Selection (3pt × 10)



1. In Cristian's Algorithm, a client sends a request at local time $T1 = 100$. The server receives the request at $T2 = 105$ and sends a response at $T3 = 110$. The client receives the response at $T4 = 125$. What is the round-trip time (δ) the client uses to estimate the clock offset?
- A. 15ms B. 20ms C. 25ms D. 30ms
- B.** Round-trip time is calculated as $\delta = (T4 - T1) - (T3 - T2)$. Here, $(125 - 100) - (110 - 105) = 25 - 5 = 20\text{ms}$.



Question Set 2: Multiple Choice Selection (3pt × 10)



2. A Chord P2P system uses an identifier space of $m = 4$ bits (16 possible IDs). Node $N = 5$ is calculating its Finger Table. What is the target ID that node $N = 5$ will look for to fill its 2nd finger table entry (where $i = 2$)?
- A. 6 B. 7 C. 9 D. 13



Question Set 2: Multiple Choice Selection (3pt × 10)



2. A Chord P2P system uses an identifier space of $m = 4$ bits (16 possible IDs). Node $N = 5$ is calculating its Finger Table. What is the target ID that node $N = 5$ will look for to fill its 2nd finger table entry (where $i = 2$)?

A. 6

B. 7

C. 9

D. 13

B. The target ID (separator) for the i th finger entry is $(n + 2^{i-1}) \bmod 2^m$. For node $N = 5$ and entry $i = 2$, the calculation is $(5 + 2^{2-1}) \bmod 16 = (5 + 2) \bmod 16 = 7$.

Question Set 3: Short Analysis (5pt × 5)



1. Processes P1 and P2 observe the following sequence: P1 writes $x=5$. P2 then reads $x=5$ and subsequently writes $y=10$. If a third process P3 reads $y=10$ and then reads $x=0$, is Causal Consistency violated? Justify your answer by identifying the happens-before relationship between the writes.

Question Set 3: Short Analysis (5pt × 5)



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Answer: Yes, Causal Consistency is violated. Because P2 read $x=5$ before writing $y=10$, there is a happens-before relationship: $W(x=5) \rightarrow W(y=10)$. Since P3 observed the effect of the second write ($y=10$), it is required to observe all causal predecessors of that write. Reading $x=0$ instead of $x=5$ contradicts this causal order.



Question Set 3: Short Analysis (5pt × 5)



2. In the context of the Paxos consensus protocol, briefly explain the difference between the “Acceptor” role and the “Learner” role. Is it possible for a single physical server node to perform both roles simultaneously?



Question Set 3: Short Analysis (5pt × 5)



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Answer: The Acceptor role is responsible for receiving and potentially accepting proposals to reach a majority, while the Learner role is passive and waits to discover what value was ultimately chosen. Yes, it is common and possible for a single physical server node to perform both roles simultaneously (often playing the Proposer role as well).



Question Set 4: System Calculations (5pt × 3)



1. You are designing a Raft-based distributed key-value store for a client who requires the system to remain available even if any 2 nodes in the cluster fail at the same time. Calculate the minimum number of total nodes (N) required for this deployment and identify the minimum number of nodes required to form a majority.



Question Set 4: System Calculations (5pt × 3)



1. You are designing a Raft-based distributed key-value store for a client who requires the system to remain available even if any 2 nodes in the cluster fail at the same time. Calculate the minimum number of total nodes (N) required for this deployment and identify the minimum number of nodes required to form a majority.

Answer: Minimum total nodes $N = 5$. Minimum number of nodes for a majority = 3. Consensus systems require $N = 2f + 1$ nodes to tolerate f failures; to survive 2 failures, the system needs $2(2) + 1 = 5$ nodes, with a majority defined as $f + 1 = 3$.



Question Set 5: System Design (5pt × 2)



1. Describe the “Prepare” phase of the Two-Phase Commit (2PC) protocol. Specifically, what action must a participant take regarding its local state before sending a **VOTE_COMMIT** (Yes) response to the coordinator, and why is this step essential for ensuring atomicity?



Question Set 5: System Design (5pt × 2)



1. Describe the “Prepare” phase of the Two-Phase Commit (2PC) protocol. Specifically, what action must a participant take regarding its local state before sending a **VOTE_COMMIT** (Yes) response to the coordinator, and why is this step essential for ensuring atomicity?

Answer: Before sending **VOTE_COMMIT**, a participant must record the transaction and its **Prepared** state to stable storage (non-volatile disk). This is essential for atomicity because it ensures that if the node crashes, it can recover and honor the coordinator’s final decision to either commit or abort, preventing the system from reaching an inconsistent state where some nodes committed and others did not.

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